Task 5

R code:

```
install.packages("vegan")

library(vegan)

# Read dataset

data <- read.table("/Users/apple/Downloads/INRTU courses/2nd semester/Data
Analysis/Homework 5/data.txt", header = TRUE, sep = "\t", check.names =
FALSE)

data

rownames(data) <- data[, 1]

data <- data[, -1]
```

1. NMDS with Bray-Curtis distance

R code:

```
# metaMDS runs multiple random starts and picks the best (lowest stress)
solution.

set.seed(42)

ord <- metaMDS(data, distance = "bray", trymax = 100)



cat("NMDS stress:", round(ord$stress, 4), "\n")
```

2. Fit significant species vectors (envfit, p <= 0.05)

R code:

```
# envfit projects each species as a vector onto ordination space

# and tests significance via permutation.

set.seed(42)

fit_sp <- envfit(ord, data, perm = 999)



# All species results (r2 and p-value)

print(fit_sp)
```

```r
# Only significant species (p <= 0.05)
# Result from this dataset: only S. cornuta is significant (p = 0.033)
cat("\nSignificant species (p <= 0.05):\n")
print(fit_sp, p.max = 0.05)
```

3. UPGMA hierarchical clustering with Bray-Curtis

R code:

```r
d_bray <- vegdist(data, method = "bray")
fit_clust <- hclust(d_bray, method = "average")  # UPGMA

# Plot dendrogram
plot(fit_clust, hang = -1,
     main = "UPGMA Dendrogram (Bray-Curtis)",
     xlab = "Sites", sub = "", ylab = "Bray-Curtis dissimilarity")
```

4. Cut dendrogram into 2-3 clusters

R code:

```r
k <- 3
clusters <- cutree(fit_clust, k = k)

# Safety fallback: reduce to k = 2 if any cluster has only 1 site
if (any(table(clusters) < 2)) {
  k <- 2
  clusters <- cutree(fit_clust, k = k)
  message("Reduced to k = 2: one cluster had only 1 site.")
}

cat("\nCluster membership (k =", k, "):\n")
```

```
print(clusters)

cat("Cluster sizes:\n")

print(table(clusters))
```

5. a) Shaded confidence ellipses (1 SD) per cluster

R code:

```
ordiellipse(ord, groups = clusters, kind = "sd", lwd = 2,

            col    = palette_cols[1:k],

            draw   = "polygon",

            alpha  = 40,

            border = palette_cols[1:k])
```

b) Site points coloured by cluster

R code:

```
points(ord, display = "sites", pch = 21, cex = 2.5, lwd = 2,

       bg  = adjustcolor(site_cols, alpha.f = 0.8),

       col = site_cols)
```

c) Significant species arrows only (p <= 0.05)

R code:

```
plot(fit_sp, p.max = 0.05, col = "darkgreen", cex = 0.85, add = TRUE)
```

d) Non-overlapping site labels

R code:

```
orditorp(ord, display = "sites", cex = 0.85, col = "black",

         priority = rowSums(data), air = 1.2)


# Legend

legend("topright",

       legend = paste("Cluster", 1:k),

       pch    = 21,

       pt.bg  = palette_cols[1:k],
```

```
      col    = palette_cols[1:k],
      pt.cex = 1.8, bty = "n")
```

6. PERMANOVA — test differences between clusters

R code:

```
site_df <- data.frame(cluster = factor(clusters))


set.seed(42)
perm_result <- adonis2(data ~ cluster,
                       data         = site_df,
                       method       = "bray",
                       permutations = 999)
print(perm_result)
```

Extract key statistics

R code:

```
# Extract key statistics
R2 <- round(perm_result$R2[1], 4)
pv <- perm_result$`Pr(>F)`[1]
```

Conclusion

The PERMANOVA result is not statistically significant ($p = 0.067 > 0.05$). Therefore, we cannot formally reject the null hypothesis that the clusters do not differ in species composition.

However, the R^2 value of 0.6439 indicates that cluster identity explains 64.4% of the total variance in Bray–Curtis dissimilarities among sites. This is a large and ecologically meaningful effect size. The lack of statistical significance is almost certainly a consequence of the extremely small sample size ($n = 6$ sites), which limits the maximum number of possible permutations to 720. With so few permutations, the minimum achievable p-value is approximately $1/720 \approx 0.0014$, making it very difficult to reach significance even when the effect is real. This is a well-known limitation of PERMANOVA with small datasets, not evidence that the clusters are ecologically meaningless.